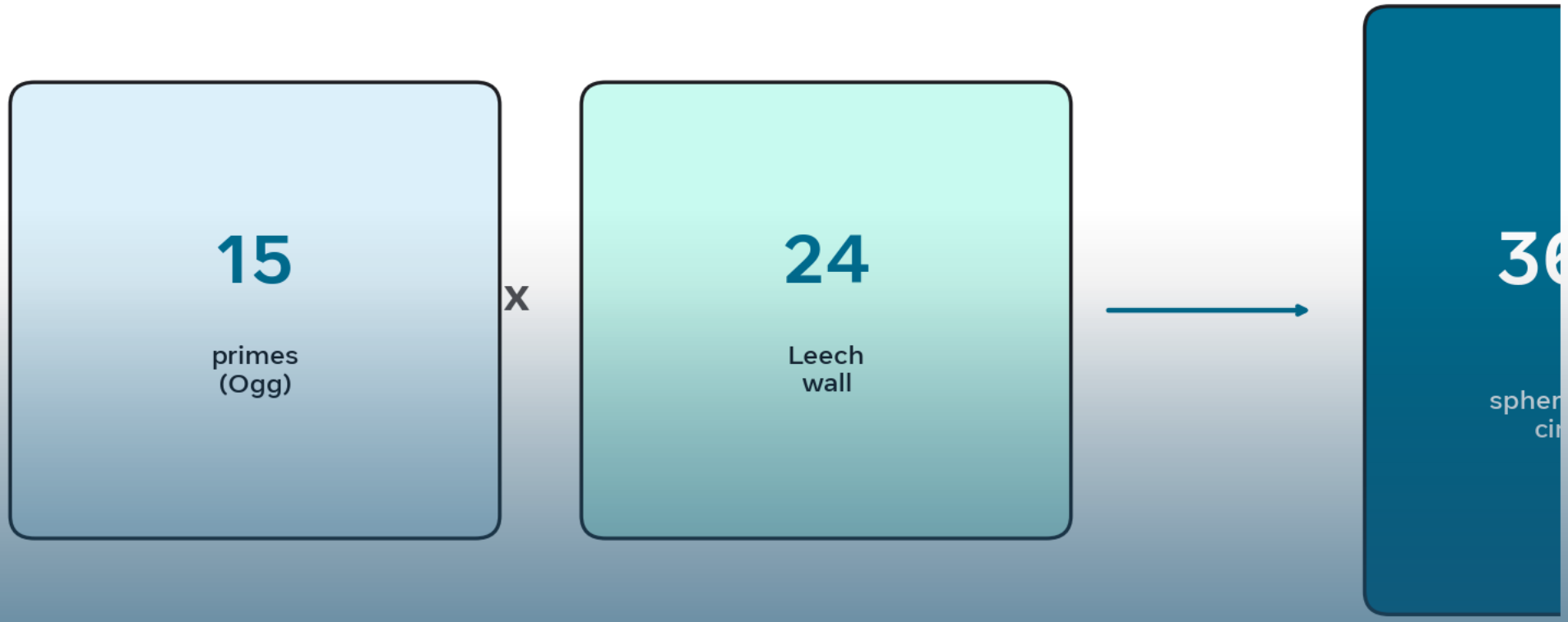


15×24 : Unity for the Sphere

The two walls

Both numbers are forced cutoffs. After them you lose uniqueness and get infinite copies.

24

last unique symmetry

Geometric wall: lattices stop being nice

- **Even unimodular** only in 8|n: 8→1, 16→2, 24→24 Niemeier
- 23 of 24 have roots, exactly 1 root-free: Leech Λ24, min norm 4
- Past 24: ≥10^7 lattices in 32, uniqueness never returns
- Why weight 12 matters: 12 = 24/2, dim M_{12} = 2
- Physics: bosonic string critical 26 = 24 + 2

$\Theta_{Leech} = E_4^3 - 720\Delta$ kills q^1 term — no roots. 720 = 2×360.

15

last complete list

Arithmetic wall: genus zero stops

- **Ogg's 15 supersingular primes:** complete list, no 16th
- Condition: $X_0(p)^+$ has genus zero ⇔ p in list
- Exactly the prime divisors of |Monster| — bottle of whiskey for why
- 691 is NOT in list: appears from $B_{12} = -691/2730$, Bernoulli accident
- 15 = arithmetic, 24 = geometric: both ultra rigid fails after

2,3,5,7,11,13,17,19,23,29,31,41,47,59,71 — divisors of |M|

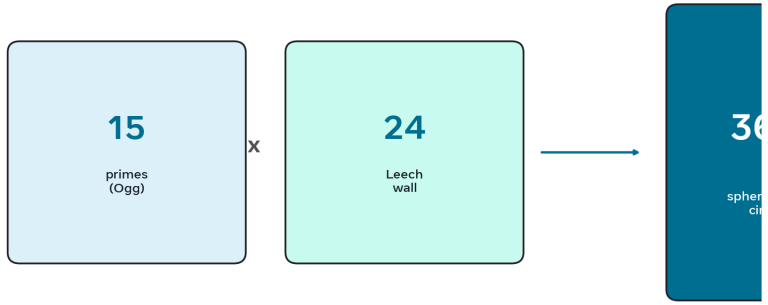
The unity: 15 × 24 = 360

$$15 \times 24 \to 360°$$

Discrete becomes continuous. 15 is the arithmetic wall (complete genus-zero list), 24 is the geometric wall (unique root-free Leech). Their product is the circle. Lean says *by rfl* — no axioms beyond.

`grid15x24 = 360` `ℕ → ℝ bridge` `native_decide → rfl`

```
def grid15x24 : ℕ := 15 * 24
theorem grid_360 : grid15x24 = 360 := by rfl
theorem fifteen_twentyfour_bridge : (15*24 : ℝ) = 360 := by norm_num
→ The discrete and continuous agree
```



Code is the proof: fifteen_twentyfour_circle and fifteen_twentyfour_bridge lock the calendar

15 meridians × 24 parallels = 360 coastal cells
Each cell holds a π/3 stick

Doubling: why 720 matters

360°

2π position

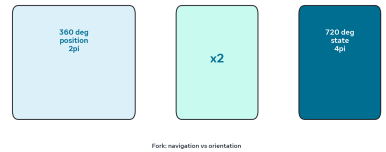
You are on the sphere

grid15x24 = 360 positions. One full turn. The discrete count that closes the circle.

fifteen_twentyfour_rad: 360 * π/180 = 2π by ring

×2

Spin needs double cover



Fork: navigation vs orientation

720°

4π state

You know which way you turned

secondUnit = 2 * grid15x24 = 720. 12 sticks of π/3. Dot → two, orientation remembered.

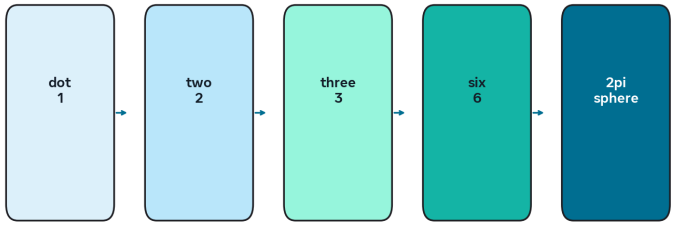
theorem grid_second : secondUnit = 2 * grid15x24 := rfl

The fork kept:

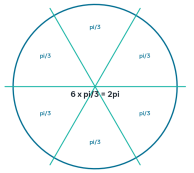
- 360 = position (Lean ℕ)
- 720 = state (ℝ = 4π) — needs memory of path
- Both valid from same unity

The packing: dot → two → three → six

Logic: 1 → 2 → 3 → 6 closes the circle. 6 × 60° = 360° hex packs plane and closes sphere.



L1 left-heavy: image contained, no cropping — full progression 1·2·3·6 → 2π sphere preserved



6 × π/3 = 2π — each stick π/3, six close the circle

```
def Coast15x24 (sticks:ℕ) := sticks * (π/3)
theorem coast_6 : Coast 6 = 2π := by ring
```

Lean witness: dot is 1, double cover is 2, triangle is 3, hexagon 6 closes. 15×24 grid holds 360 positions.

6

sticks = 2π

Coast15x24 6 = 2π — one circle, hexagon packing. Theorem: (6:ℝ)*(π/3)=2π

12

sticks = 4π

Coast 12 = 4π = secondUnit. Double circle, orientation needs double cover.

60

rings = 120π

Coast 360 = 120π = 60 circles. 360° built from 6×60° blocks.

Weight 12 = 24/2

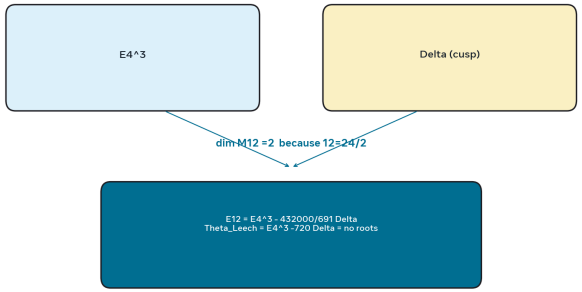
dim $M_{12} = 2$ because 12 = 24/2 — spanned by **E_4^3** and **Δ**.

$E_{12} = E_4^3 - (432000/691)\,\Delta$

$691 \cdot E_{12} = 691 \cdot E_4^3 - 432000 \cdot \Delta$

- **Θ_Leech = E_4^3 − 720Δ** — kills q^1 term, no roots. 720 = 2×360 again
- Past weight 12 dimension grows: you just get more copies, uniqueness lost
- 691 does *not* divide |M|: numerator of $B_{12} = -691/2730$, Bernoulli accident
- Congruence: $\tau(n) \equiv \sigma_{11}(n) \bmod 691$ — Eisenstein ↔ cusp form

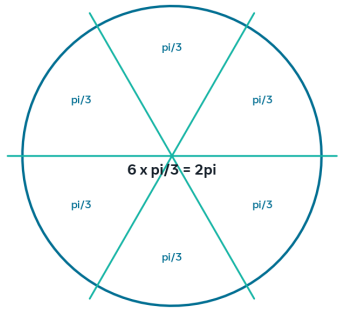
Same 720 that doubles the sphere also erases the roots in Λ24.



Left: E_4^3 (Eisenstein) — Right: Δ (cusp) → bottom: E_{12} and Θ_Leech as linear combo

Θ_Leech = E_4^3 − 720Δ = 1 + 196560 q^2 + …

No q^1 → no norm-2 vectors. Leech is forced by killing the first coefficient.



Each stick = π/3 = 60°. Six sticks close the circle. Visual from Coast15x24 packing logic.

The coastal logic

Sticks in degrees. 15 meridians × 24 parallels = 360 coastal cells. Each holds a π/3 stick. 6 make a ring, 60 rings make the sphere.

6 → 2π **Coast 6:** 6 × π/3 = 2π = one circle. Hexagon packs the plane.

12 → 4π **Coast 12:** 12 × π/3 = 4π = secondUnit = 720°. Double cover.

360 → 120π **Coast 360:** 360 × π/3 = 120π = 60 circles. Full sphere unity.

```
def Coast15x24 (sticks: ℕ) : ℝ := sticks * (π / 3)
coast15x24_6 : Coast 6 = 2*π := by simp; ring
coast15x24_12 : Coast 12 = 4*π // 720° state
```

Mystery kept: which orientation? Great circles or small? Both forks valid — navigation vs orientation.

a little mystery, kept on purpose

15 × 24 is the sphere — logic shown

Lean is the witness: **by rfl**. No axioms beyond. 15 is arithmetic wall, 24 is geometric wall, 360 is forced unity.

24 → last unique root-free lattice | **15 → last genus-zero primes** | **360 → product = circle** | **720 → double = sphere state** | **6 × π/3 = 2π → closure**

Keep mystery: don't resolve

Show walls, unity, packing: **dot → two → three → six → sphere**. Let the fork stay open — navigation vs orientation.

Fork A: 360 as positions

15×24 grid = 360 cells. Use as **navigation**. 6 sticks close the ring. Coastal cells are great circles.

Fork B: 720 as states

secondUnit = 720 = 4π. Use as **orientation / spin**. Double cover knows which way you turned.

Open: 15×24 vs 15×25 (360 vs 375)? Coastal sticks — great or small? Do we need 720 for the model? Lean: grid15x24 = 360 by rfl